

Cluster Sampling Design Learning Activity

STA304 — Summer 2026

Learning Activity 8 Overview

A school district wants to estimate the **average number of hours per week** students spend on homework. The district has $N = 5$ classrooms (the primary sampling units, or PSUs), each containing $M = 6$ students. Rather than surveying every student, you will design a **two-stage cluster sample**: first randomly select n classrooms, then randomly subsample m students from each selected classroom.

A pilot survey conducted last semester provides estimates of between- and within-cluster variability, and you have been given a survey budget. Your job is to choose n and m to minimise variance subject to the cost constraint, and to ensure the final estimate meets a desired level of precision.

Background

Population data

The full population of 30 students and their weekly homework hours (from administrative records, used here to let you verify your estimates) is shown in Table 1.

Table 1: Weekly homework hours for all 30 students across 5 classrooms

Classroom (PSU)	S1	S2	S3	S4	S5	S6	Room mean
Room 1	4	6	5	7	3	5	5.00
Room 2	8	9	7	6	8	10	8.00
Room 3	2	3	4	2	5	3	3.17
Room 4	6	5	7	4	6	8	6.00
Room 5	10	9	11	8	10	12	10.00
Population mean	$\hat{y}_U = 6.43$ hrs/week						

Pilot survey estimates

From a small pilot survey conducted last year, the following mean squares have been estimated:

$$\widehat{\text{MSB}} = 14.2 \quad \widehat{\text{MSW}} = 2.8$$

where MSB is the mean square **between** classrooms and MSW is the mean square **within** classrooms.

Variance formula

The variance of the cluster sample mean $\hat{y}$ under two-stage sampling with equal PSU sizes is:

$$V(\hat{y}) = \underbrace{\left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM}}_{\text{Term 1: between-cluster}} + \underbrace{\left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}}_{\text{Term 2: within-cluster}} \quad (1)$$

Cost model

Each classroom visited incurs a fixed travel and setup cost, and each student interview has a per-unit cost. The total survey cost is:

$$\text{Cost} = c_1 \cdot n + c_2 \cdot n \cdot m \leq C \quad (2)$$

with $c_1 = 4$ (cost per PSU visited), $c_2 = 1$ (cost per student interviewed), and total budget $C = 20$.

Part A: Enumerate Feasible Designs

- A. Using the cost constraint (2), derive an expression for the maximum affordable m as a function of n :

$$m_{\max}(n) = \min(6, \lfloor \dots \rfloor)$$

- B.** Complete Table 2 by computing $m_{\max}$, total cost, and the two variance terms for each valid value of n . Show at least one row of working below the table.

Table 2: Feasible two-stage designs under the cost constraint

n	$m_{\max}$	Cost	Term 1	Term 2	$V(\hat{y})$
1					
2					
3					
4					
5	<i>(is this feasible?)</i>				

- C.** Which design minimises $V(\hat{y})$? State your chosen (n^*, m^*) .

Part B: Precision Determination

A researcher typically specifies a desired **margin of error** d (the half-width of a 95% confidence interval) and finds the minimum n such that the variance is small enough:

$$V(\hat{y}) \leq \left(\frac{d}{1.96} \right)^2$$

For each n , set $m = m_{\max}(n)$ to make best use of the budget.

- A. The district requires that the estimate be within ± 2.0 hour per week (95% CI). What is the required upper bound on $V(\hat{y})$?

$$V(\hat{y}) \leq \left(\frac{2.0}{1.96} \right)^2 =$$

- B. Using your completed Table 2, identify the minimum n that achieves this precision. Does your chosen design from Part A meet this requirement?

- C. Now suppose the district tightens its requirement to ± 1 hours. Repeat the calculation. Is this achievable within the budget?

$$V(\hat{y}) \leq \left(\frac{1}{1.96} \right)^2 =$$

Part C: Conduct the experiment

- A. Use a random number generator to draw a sample according to your chosen sample from Part A. Which PSUs and SSUs did you sample?

- B.** Using the survey package or by hand, estimate the population mean and a 95% confidence interval. Use t_{n-1} for your critical values.

- C.** Is the population mean in your confidence interval? Is the margin of error within the desired level of precision? Comment on any discrepancies.